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Mapping bull kelp canopy in northern California using Landsat to enable long-term monitoring

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ABSTRACT

Extending from central California to Alaska, bull kelp (*Nereocystis luetkeana*) forms seasonal kelp forests that are iconic coastal ecosystems in much of the eastern Pacific. Historical and ongoing field surveys and aerial imagery are used to provide biological data on kelp canopy cover and health, but satellite remote sensing provides the opportunity to generate consistent, long-term datasets over a large spatial scale. Robust satellite-based timeseries measurements of giant kelp (*Macrocystis pyrifera*) are available for much of the California coastline (e.g., Santa Barbara to Santa Cruz), but there have been no equivalent publications for bull kelp. Recent loss of bull kelp in northern California emphasized the need for more expansive long-term monitoring of canopy trends. We tested various kelp classification approaches using Multiple Endmember Spectral Mixture Analysis (MESMA) applied to Landsat imagery, which allowed sufficient temporal and spatial data collection during bull kelp's narrow seasonal maximum, and compare with the California Department of Fish and Wildlife (CDFW) aerial survey canopy area product. We addressed five main topics that have relevance to enabling Landsat in long-term monitoring of bull kelp canopy coverage in northern California: (1) the effect of MESMA configurations applied to Landsat imagery, including software dependencies and endmember configurations, (2) comparison of Landsat to traditional surveys, (3) differences across Landsat sensors, (4) tidal influence on canopy area, and (5) trends in the decadal timeseries. We found that there was no statistical difference ($p = 0.53$) between MESMA platforms (IDL-based and a Python-scripted method; RMSE 1.5 km^2 or 17.6% when normalized by the range in CDFW), and that a 7-endmember MESMA model provided the lowest RMSE (1.4 km^2 or 16.9%). Furthermore, while tidal phases can submerge or emerge kelp canopy and thus potentially affect kelp detection, we found a weak and statistically insignificant correlation between tides and performance of our remote estimate of kelp canopy area. Canopy estimations from Landsat-8 images yielded a higher NRMSE than Landsat-4 and -5 images, but the lack of matchup limits comparison. Imagery from early fall yielded the largest coverage estimates. Overall, our results show that Landsat enables broad remote measurement of bull kelp canopy coverage to supplement existing survey methods and increase continuity of timeseries for monitoring long-term trends.

1. Introduction

Bull kelp (*Nereocystis luetkeana*) is a brown macroalgal species that forms the iconic kelp forests present in northern California and throughout the western coast of North America (Springer et al., 2010). It anchors to rocky substrates in temperate coastal waters and, as a foundation species, supports a complex nearshore ecosystem. Thus, bull kelp forests provide essential ecosystem services directly benefitting coastal communities through support of economically important marine fisheries and recreational opportunities (Springer et al., 2010), reduction in nearshore wave strength (Løvås and Tørum, 2001), and historical kelp harvesting for the Pomo Native Americans (Fredrickson and Peri, 1985). However, the regional extent of bull kelp forests can be reduced due to changes in oceanic conditions (Pfister et al., 2017; Schiel et al., 2004; Springer et al., 2010; Thomsen et al., 2019). Bull kelp has largely been able to recover quickly following these conditions (Tegner and Dayton, 1987), but a recent large-scale bull kelp decline in northern California, which was concurrent with a marine heat wave and strong El Niño event, suggests that the ecosystem's resilience to large physical disturbances was low and highlights the need to monitor the species over an

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appropriately long time span (McPherson et al., 2020; Rogers-Bennett and Catton, 2019).

Remote sensing, in the form of satellite and aerial imagery, has emerged as an effective means to map and monitor kelp forests. Satellite remote sensing methods are more cost- and time-efficient than boat and diver surveys for kelp (Cavanaugh et al., 2010), allowing for easily replicated canopy surveys covering broader time periods and spatial scales (Schroeder et al., 2019). Various remote sensing methods have been used to map another canopy-forming kelp species, giant kelp (*Macrocystis pyrifera*), along the western coastline of the United States (e.g., Augenstein et al., 1991; Bell et al., 2020; Cavanaugh et al., 2010) and to map combined canopy for both giant and bull kelp (Nijland et al., 2019; Pfister et al., 2017). Fewer efforts, though, have focused on bull kelp, and these have been limited to North America's Pacific Northwest (Hamilton et al., 2020; Schroeder et al., 2019; Stekoll et al., 2006).

Despite sharing canopy-forming morphologies, there are a number of differences between the two kelp genera that lead to unique challenges when attempting to develop a satellite remote sensing effort for bull kelp compared to giant kelp. Both genera are found throughout the California coastline and co-occur in many areas, but giant kelp is largely absent in northern California, in part because of its lower tolerance to energetic wave action (Foster et al., 2013; Schroeder et al., 2019; Springer et al., 2010). Bull kelp is an annual kelp species, regenerating its entire biomass and canopy every year. Giant kelp, as a perennial species, is capable of retaining biomass from year to year in the absence of physical and biological perturbations (Springer et al., 2010). Both form canopies composed of leaf-like blades, but giant kelp blades largely float at or above the ocean surface while bull kelp blades may remain suspended in a subsurface layer below the floating stipe and pneumatocyst (Fig. 1; Jensen and Estes, 1980; Koehl, 1984; Schroeder et al., 2019). Bull kelp remote sensing must thus account for natural variability in distribution, phenology, and vertical position that differ from those of giant kelp.

The California Department of Fish and Wildlife (CDFW) has collected aerial imagery of kelp beds to map canopy coverage, but the state-funded flights are resource-intensive and thus cannot be undertaken annually (Aerial Kelp Surveys, 2019; Jensen and Estes, 1980; Veisze et al., 1999). Other attempts have been made to map small extents of bull kelp canopy through private satellites and aerial photography (O'Neill and Costa, 2015; Schroeder et al., 2019; Stekoll et al., 2006), but our work focused on mapping a large area of bull kelp canopy with publicly available satellite data in order to enable large-scale sustained and historical observations. Landsat, a satellite system dating back to 1972 that is operated by the U.S. Geological Survey (USGS), produces images covering the entire Earth every 16 days, allowing for long-term monitoring with consistent data (Williams et al., 2006). The 1982 introduction of the Thematic Mapper (TM) onboard Landsat-4 (LS-4)

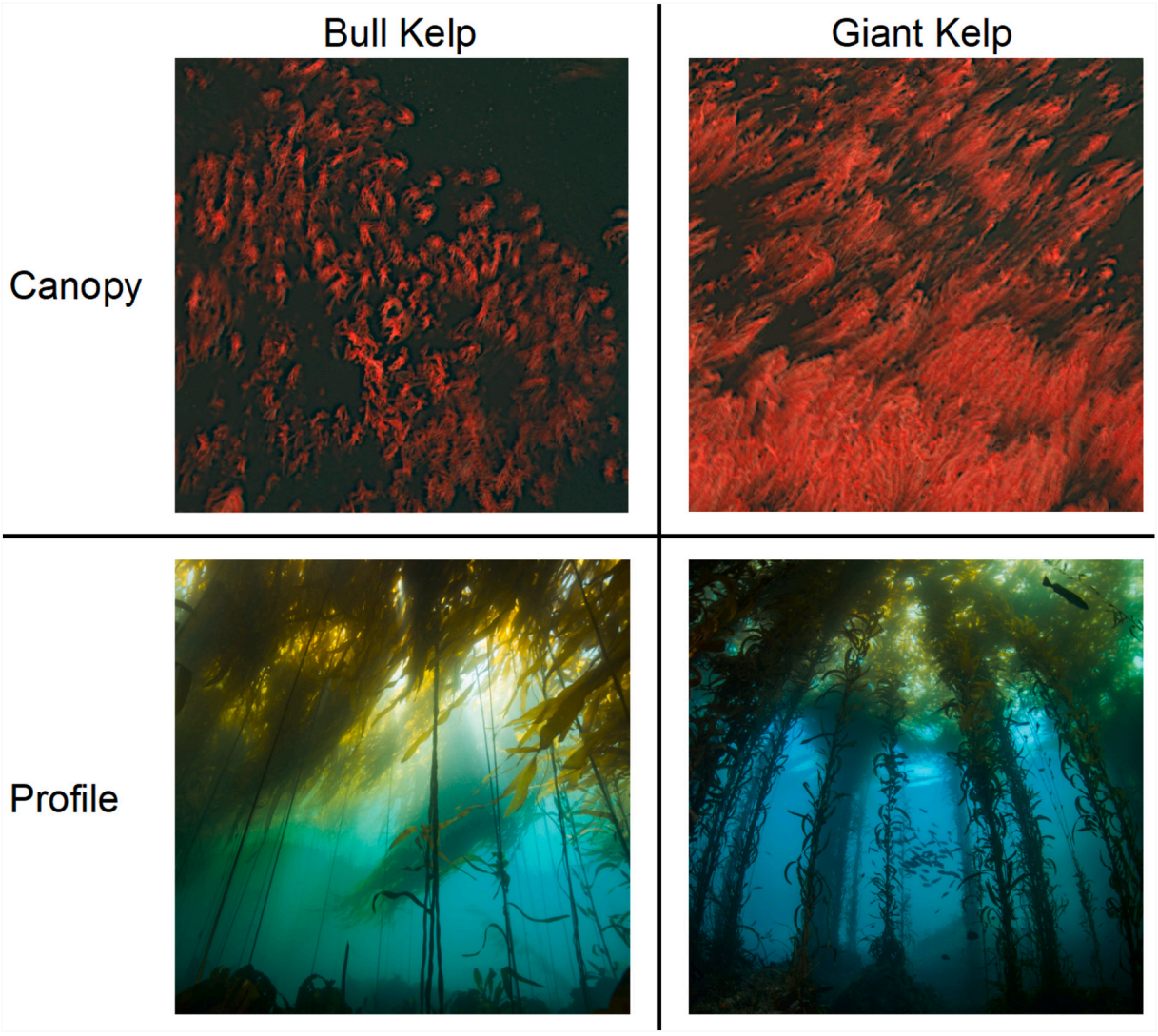


Fig. 1. Comparison of giant kelp and bull kelp surface canopy (false color) and profile images. In situ images were courtesy of Michael Langhans (UCSC). Canopy images were obtained by M. McPherson with a multispectral RedEdge-M camera (MicaSense) mounted on a DJI Matrice 100 quadcopter.

provided relatively high spatial resolution (30 m) multispectral imagery, which has enabled detection of kelp from successive Landsat platforms through the current sensor, LS-8 (Bell et al., 2020).

Past efforts to estimate kelp canopy have used Normalized Difference Vegetation Index (NDVI), a tool applied primarily in terrestrial environments that utilizes the difference in reflectance between near-infrared and red spectral bands (Cavanaugh et al., 2010; Stekoll et al., 2006). For this analysis, we instead chose to use Multiple Endmember Spectral Mixture Analysis (MESMA), an image classification method that utilizes a multispectral bandset derived from TM (compared with the two-band approach for NDVI; Stekoll et al., 2006) and enables classification of mixed pixels (Roberts et al., 2017). Recent research has applied MESMA to estimate giant kelp biomass in southern California (Bell et al., 2020) and southern South America (Friedlander et al., 2020) and bull kelp biomass in Oregon (Hamilton et al., 2020), providing a template for a similar analytical approach applied to bull kelp in northern California. Here, we applied MESMA to Landsat satellite imagery in order to estimate the canopy area of bull kelp forests. The goal of this study was to develop a robust remote sensing tool to accurately measure bull kelp canopy area in northern California using Landsat imagery by (1) comparing various data processing pathways and implementations of MESMA (i.e., the number of endmembers, spectral shape of the kelp endmember, and software platform) on the performance of the kelp canopy coverage estimates, (2) analyzing differences in canopy estimates between relatively low spatial resolution satellite data versus relatively high spatial resolution aerial surveys, (3) comparing performance between Landsat sensors, (4) evaluating the influence of tidal height on Landsat-derived canopy estimation, and (5) investigating bull kelp canopy climatology in the context of seasonal kelp detection.

2. Methods

2.1. Study area

We evaluated Landsat imagery from within a region of interest in northern California that spans from Point Arena to Jenner (latitude 38.95°N to 38.40°N; Fig. 2), which was surveyed by CDFW irregularly during our timeseries and which encompasses some of the area where the recent bull kelp canopy decline has been described. The region covers more than eighty-five kilometers of coastline (Landsat scene ID: 045033; S1) and solely or predominantly contains bull kelp (Springer et al., 2010), and thus we assumed giant kelp to be absent.

2.2. Data assemblage

We downloaded Landsat Collection 1 Level-2 satellite imagery from the USGS Earth Explorer portal (Earth Explorer, 2019; S1). We chose 88 cloud-free images collected predominantly during periods of bull kelp maximum canopy biomass from August through November (Springer et al., 2010). For a separate climatological analysis (Section 3.4), we also selected multiple images throughout the same year.

The region's coastlines were masked in L3Harris Geospatial's ENVI program by manually tracing the coastline from a Landsat image with a buffer to explicitly include ocean without any land or nearshore swell (Hamilton et al., 2020; Fig. 3); this buffering may exclude some coastal kelp but has been shown to increase detection accuracy (Nijland et al.,

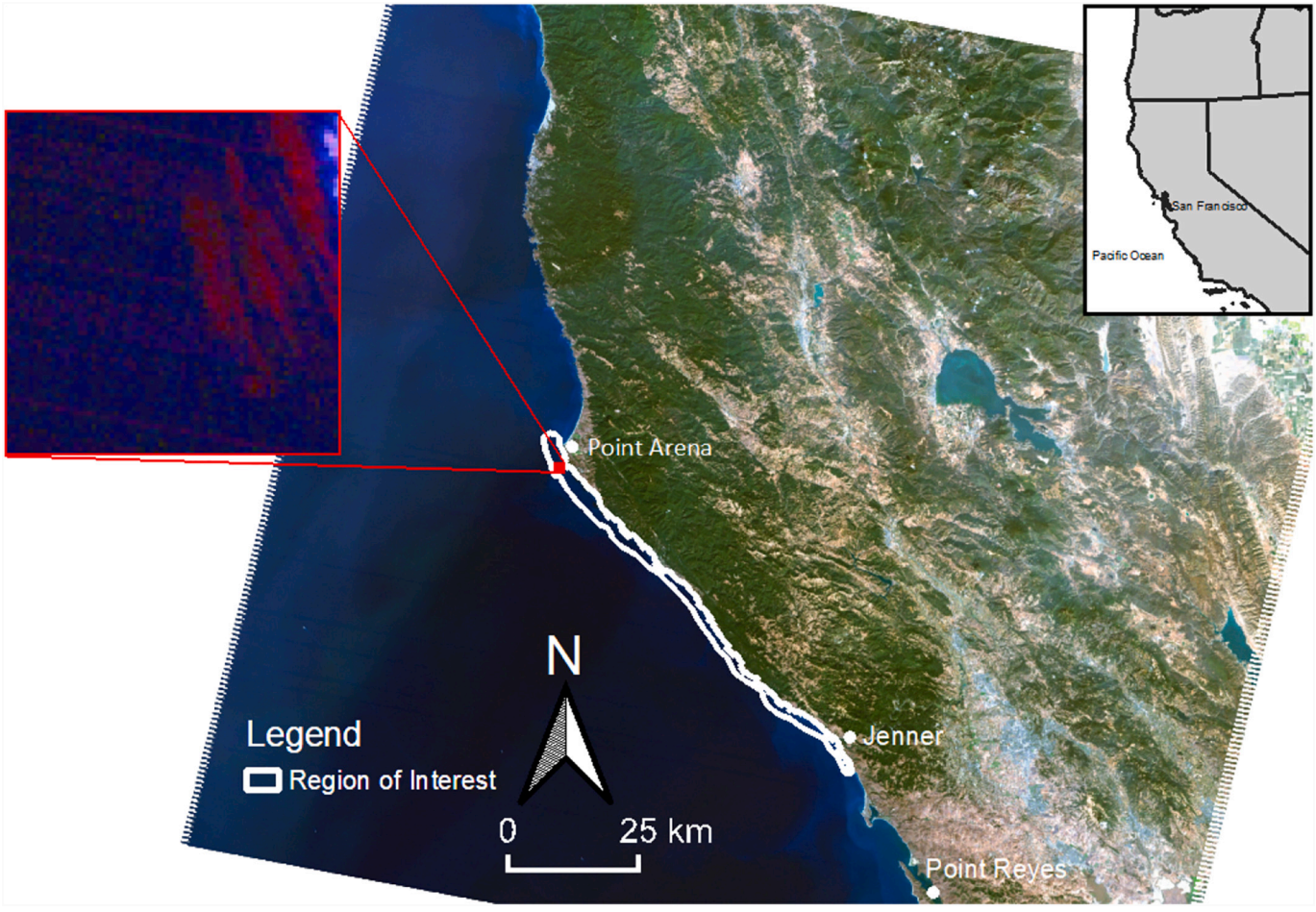


Fig. 2. Landsat satellite image showing the region of interest, with an inset map showing location within northern California and a false-color image of kelp beds in Point Arena.

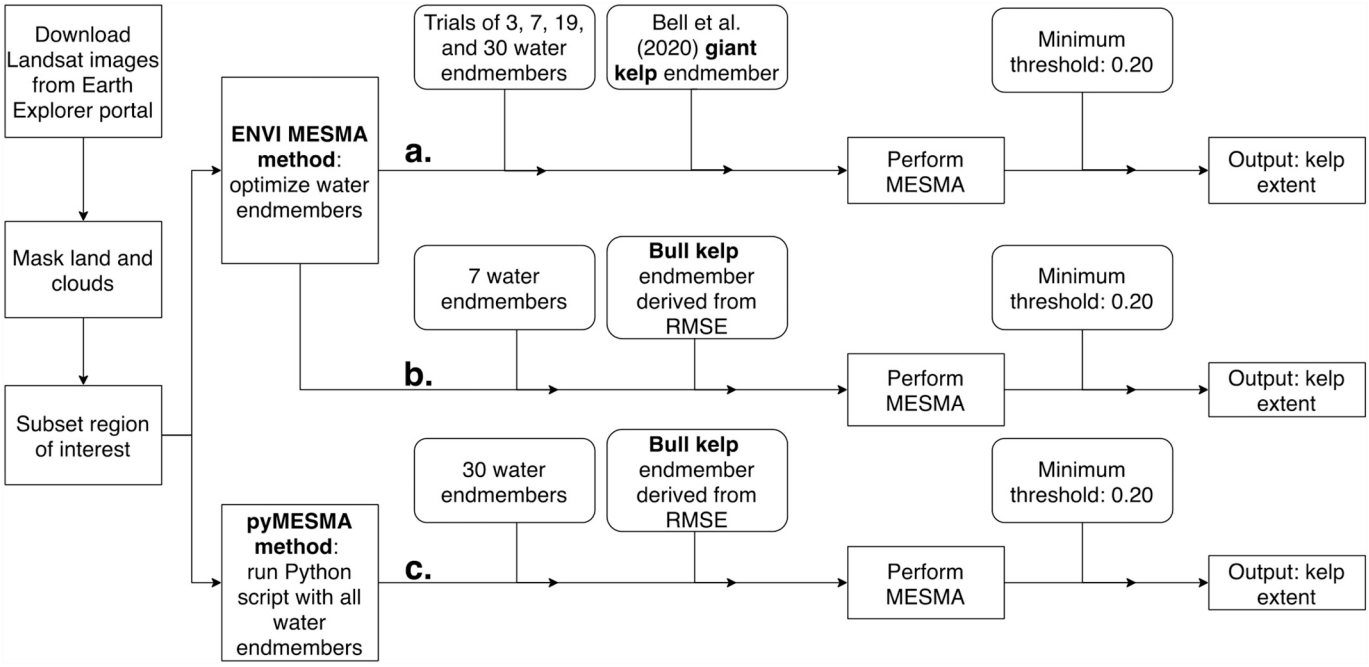


Fig. 3. Flow chart outlining our basic methodology, beginning with downloading and subsetting the Landsat dataset and progressing to our MESMA trials using kelp and seawater endmembers. a. ENVI MESMA method with a giant kelp endmember. b. ENVI MESMA with a bull kelp endmember. c. PyMESMA with a bull kelp endmember.

2019). Certain images contained small clouds, which we manually excluded using a built-in ENVI masking function. Images were subset to the area of interest and spectral data for Landsat were exported for blue (0.45–0.52 μm in LS4/5, 0.45–0.51 μm in LS8), green (0.52–0.6 μm in LS4/5, 0.53–0.59 μm in LS8), red (0.63–0.69 μm in LS4/5, 0.64–0.67 μm in LS8), and near-infrared wavelengths (0.76–0.90 μm in LS4/5, 0.85–0.88 μm in LS8) for all images (Earth Explorer, 2019; Fig. 3).

We selected images from LS-4, −5, and − 8 satellites. LS-6 failed to reach orbit and LS-7 experienced a scan-line corrector failure prior to years that could be compared to aerial surveys (Williams et al., 2006), precluding use of those two sensors. Using the Landsat sensors, we constructed a long-term timeseries starting in 1982 which could be extended into the future as new, cloud-free Landsat imagery becomes available.

2.3. Methodological approach

MESMA is an image classification add-in tool that uses a library of spectral endmembers, or the pure spectra associated with sub-pixel scale targets, to identify and distinguish between surface features (Roberts et al., 1998). We conducted a number of trials to compare the influence that the following had on the performance of MESMA: (1) water endmember quantity, (2) giant kelp versus bull kelp endmembers, and (3) different software platforms.

2.4. MESMA endmember selection

MESMA requires pre-selection of endmember spectra for classification of an image, where endmembers are single pixel spectra representing a distinct surface feature. For this analysis, our two endmember classes were kelp and seawater, since clouds and land were pre-masked. In order to distinguish between kelp and seawater, and thus map the spatial extents, we selected multiple water endmembers and a single kelp endmember. We manually selected 30 pixels from our Landsat scene to serve as water endmembers, which included spectra from coastal water and river plume features to encompass the considerable variability in water quality and glint conditions across Landsat pixels in our study area (Bell et al., 2020; Roberts et al., 1998).

2.5. Endmember analysis

The type and number of water endmembers were optimized using MESMA in ENVI (hereafter referred to as “ENVI MESMA”) by pre-selecting a subset of the 30 representative water endmembers, described above. We used the Optimize Spectral Library function in ENVI MESMA to select endmembers from the coast, river plumes, and open ocean that had the lowest RMSE and spectral angle values (Roberts et al., 1998). For the trials using the giant kelp endmember, we ran a sensitivity analysis between MESMA trials using 3, 7, and 19 optimized water endmembers (Fig. 3a). For trials with the bull kelp endmember, we used 7 optimized water endmembers (Fig. 3b; Roberts et al., 2017; Tane et al., 2018). We also ran a 30-endmember Python-scripted approach (hereafter referred to as “pyMESMA”; Fig. 3c) and used ANOVA to determine whether there were significant differences in kelp canopy cover between ENVI MESMA and pyMESMA 7- and 30-endmember approaches; the ANOVA recorded no statistical differences between the methods (Section 3.1), prompting us to use pyMESMA data in subsequent analyses.

We performed MESMA in separate runs using two different kelp spectral endmembers: one giant kelp endmember as a convenient proxy for bull kelp (Fig. 3a; Bell et al., 2020), and another created from blue, green, red, and near-infrared wavelengths (bands 1–4 in LS-4 and -5, bands 2–5 in LS-8) from single pixels near the center of relatively large bull kelp canopies within our region of interest from five separate images. We then utilized the Endmember Average RMSE (EAR) tool in MESMA to select the single endmember with lowest Root Mean Square Error (RMSE; Fig. 3b and c; Dennison and Roberts, 2003). This approach was consistent with past studies of giant kelp (Bell et al., 2020; Cavanaugh et al., 2011).

With these endmembers, we built a spectral library and ran MESMA on all 88 of our images, yielding kelp canopy fractional coverages to which a minimum threshold of 0.20 (Hamilton et al., 2020) was applied, either with the ENVI MESMA Minimum Fraction feature or in Python (Fig. 3a–c). Thus, all pixels with kelp fractions less than the threshold

were classified as water. For those pixels with kelp, we multiplied the fractional value by the Landsat pixel size (30 × 30 m, or 900 m^2) to estimate the approximate area of kelp canopy within the pixel (Williams et al., 2006). We then summed the individual kelp pixel areas and converted to square kilometers to estimate the canopy area for our entire region of interest.

2.6. Spectral and spatial analysis

We compared the ENVI MESMA results (using both the giant kelp and bull kelp endmember) and pyMESMA results (using only the bull kelp endmember) with CDFW aerial kelp surveys. CDFW surveys utilize 2 m spatial resolution to map the kelp beds along the California coastline and were conducted nine times within our study region and during our study period (1989, 2003 to 2005, 2008, 2009, and 2014 to 2016; Aerial Kelp Surveys, 2019; S1). For each year, we took the sums of both the CDFW and MESMA total kelp areas, which provide resource managers with estimates to inform decision making, such as setting maximum harvest limits (Anderson et al., 2007). We then evaluated the correlation between CDFW and MESMA methods by calculating RMSE to represent uncertainties between their canopy areas (Bell et al., 2020). We also calculated the Normalized RMSE (NRMSE) by dividing the RMSE values by the CDFW canopy area range. We applied the coefficient of determination (R^2) to compare the agreement in canopy coverage between our derived MESMA canopy area and CDFW canopy area (Bell et al., 2020; Cavanaugh et al., 2010; Cavanaugh et al., 2011; Hamilton et al., 2020). A kelp presence/absence metric using pyMESMA data with a minimum fractional threshold of 0.20 was used when comparing Landsat estimates with CDFW surveys (Nijland et al., 2019), as this criterion aligned with the kelp identification methodology employed by CDFW on aerial survey data.

We compared the spatial distribution of bull kelp canopy detected by CDFW and pyMESMA for individual years at Point Arena and evaluated the performance using pixel-by-pixel matchups between the Landsat and CDFW datasets. The 2 m spatial resolution CDFW data were resampled to the 30 m resolution of Landsat data and fractional cover calculated as the proportion of the new 30 m pixel with kelp present (Hamilton et al., 2020; Nijland et al., 2019). An accuracy assessment of the CDFW and Landsat fractional kelp coverage was evaluated using an error matrix, an established method for comparing classifications between remote sensing data and a reference dataset; we treated CDFW aerial survey data as our reference. We then used random stratified sampling to choose 100 validation points classified as kelp by CDFW and 100 classified as water for each year (Congalton, 1991; Nijland et al., 2019; Schroeder et al., 2019), making sure that kelp points were at least 15 m apart to collect a sufficient number of points while minimizing autocorrelation (Cavanaugh et al., 2013; Hamilton et al., 2020). Our matrix presents producer's accuracy (probability that CDFW classification is correctly classified by MESMA), user's accuracy (probability that a Landsat-derived MESMA classification represents the corresponding CDFW classification), and total accuracy (correctly classified pixels divided by total kelp and water pixels from the random stratified sampling). In order to mitigate the potential effects of class imbalance on the interpretation of the confusion matrix, we also compare CDFW and Landsat direct matchups using the Matthews Correlation Coefficient (ranging from −1 to 1), which compares the positive and negative elements of a confusion matrix with less sensitivity to different numbers of elements (Thusberg et al., 2010; Vihinen, 2012). Finally, we applied RMSE statistics to compare coverage estimates between LS-4/LS-5 (data from 1989, 2003 to 2005, 2008, and 2009) and LS-8 satellites (2014 to 2016).

2.7. Tidal analysis

We investigated the potential influence of tides on bull kelp detection by using hourly tidal cycle data available from NOAA for the Point Arena subregion (Fig. 2; NOAA Tides and Currents, 2019; S1). Point Arena tidal data were unavailable from 1984 to 1990, but a Spearman's correlation analysis revealed a strong, monotonic positive correlation between the non-linear Point Arena and Point Reyes tidal datasets across 21 dates ($p < 0.001$, $r_s = 0.98$), prompting us to use the Point Reyes dataset when Point Arena data were unavailable. We compared the tidal level to pyMESMA canopy area estimates across our entire timeseries, analyzing variance with a Goldfeld-Quandt test using the gqtest function in the R-Studio package lmtest v 0.9–38, and separately compared tidal level to the normalized residuals of the pyMESMA approach (calculated as the difference between the CDFW and pyMESMA estimates) across the nine dates where CDFW surveys were available. Doing so evaluated the overestimation or underestimation of canopy area as a function of tidal height, for example due to submergence of the surface canopy with increasing tidal height, or emergence of the surface canopy with decreasing tidal height.

2.8. Climatological analysis

We compared monthly variability in canopy coverage by analyzing pyMESMA results from 88 scenes showing average monthly coverage across a 35-year timeseries and from an 18-scene subset showing month-to-month changes within single years (Section 3.4). These results enabled comparison of seasonal changes in canopy coverage over the course of the kelp growing season (early spring to late summer), which was used to evaluate the sensitivity of our detection to the seasonal cycles in bull kelp canopy extent. This analysis also informed our selection of imagery to generate the timeseries that we evaluate in Fig. 4 and the matchups with the CDFW dataset that we consider in Fig. 5.

3. Results

3.1. Endmember analysis

We used varying numbers of water endmembers to optimize ENVI MESMA by quantifying kelp area and RMSE values. Canopy areas in Fig. 4 were obtained with a kelp endmember originally used for giant kelp (Fig. 3a; Cavanaugh et al., 2011). Trials comparing 3, 7, 19, and 30 endmembers indicated that the 3- and 7-endmember models consistently produced the highest kelp coverage (Fig. 4). The 19- and 30-endmember models had relatively large NRMSE values (20.7% and 20.2%, respectively) based on nine years of CDFW kelp surveys (1989, 2003 to 2005, 2008, 2009, and 2014 to 2016), while the 3- and 7-endmember models had lower NRMSE values (18.3% and 17.9%, respectively), leading us to use 7 water endmembers in subsequent optimizations (Fig. 3b). However, our ANOVA analysis with a significance level of 0.05 showed that these four estimates of canopy coverage did not significantly differ within the imagery evaluated ($p = 0.90$; $F = 0.20$).

The bull kelp endmember we developed produced a more accurate estimation of canopy cover, with lower RMSE and higher R^2 values than those of the giant kelp endmember (Table 1). The use of a bull kelp endmember with 7 optimized water endmembers produced lower RMSE values than the pyMESMA method that used the bull kelp endmember. There was no water endmember optimization step in the pyMESMA processing, which used all 30 spectra to encompass oceanic variability. The Python scripting, however, significantly reduced processing times for both user and computer (Table 1). An ANOVA analysis on the entirety of the timeseries found that these three methods were not statistically different ($p = 0.53$, $F = 0.64$), prompting us to use pyMESMA in subsequent analyses.

3.2. Spectral and spatial analysis

Across 5 years of CDFW and Landsat matchups (Fig. 5a), patterns of canopy area were similar in distribution and temporal trends. Comparing all years with CDFW and Landsat matchups available (Fig. 5

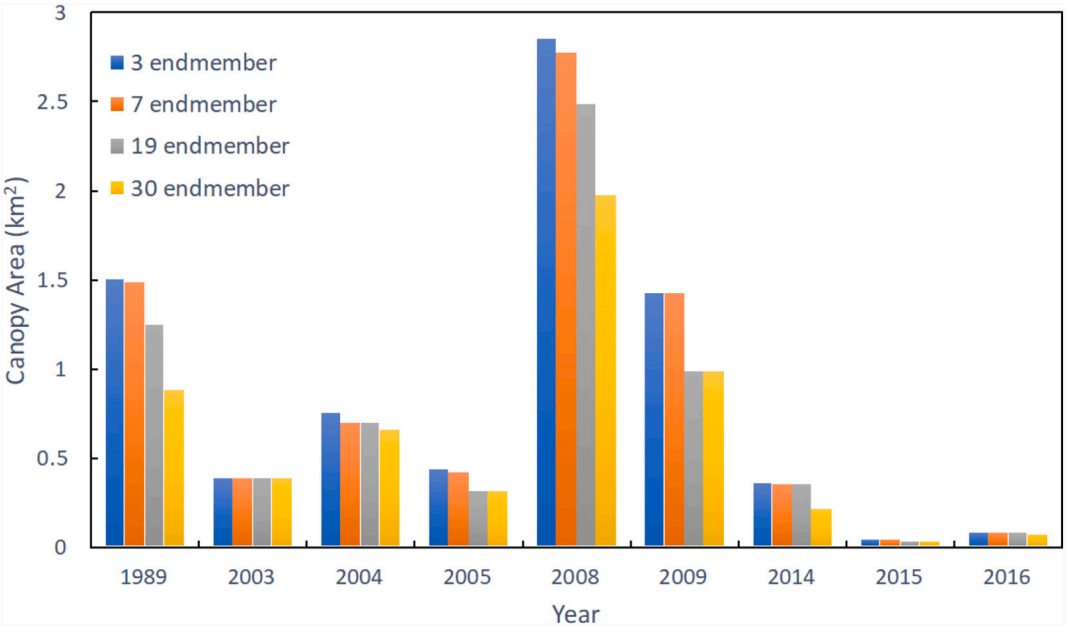


Fig. 4. Comparison of MESMA runs with varying numbers of water endmembers. The 30-endmember run utilized all 30 user-specified water endmembers, while the 19-, 7-, and 3-endmember trials used a subset of the 30 endmembers based on EMC endmember optimization.

panels b and c), there was relatively low error (NRMSE = 12.7%) and strong correlation (R^2 = 0.88) to pyMESMA presence/absence results (Fig. 5b). MESMA enabled the additional estimation of fractional canopy coverage, which resulted in a lower absolute canopy area and a slightly degraded R^2 compared with the Landsat presence/absence approach (Fig. 5c). The least squares fit between the CDFW data and the Landsat estimate was nearer a 1:1 relationship for the presence/absence approach compared to the fractional approach because the assignment of fractional coverage reduces the sum of the kelp canopy areas and because the CDFW product does not assign fractional coverages. Both MESMA and CDFW datasets distinguish high versus low kelp extents, but lower log-space uncertainties for higher coverage of bull kelp (Fig. 5 panels b and c) suggest that differences between the two methods were most pronounced when kelp canopy was low (2014 to 2016). This is in agreement with previous findings suggesting that Landsat sensitivity and accuracy are reduced for patches that contain less than 20% kelp canopy biomass (Hamilton et al., 2020), corresponding to 180 m^2 of kelp in a 900 m^2 Landsat pixel if the kelp were spatially continuous. Furthermore, the pyMESMA coverage derived from Landsat exhibited an expanded range (approximately three orders of magnitude) between high (1989, 2003 to 2009) and low biomass years (2014 to 2016). However, we note that CDFW surveys capture finer scale spatial variability than is achieved using the Landsat methods (Fig. 5a).

Our accuracy assessment between fractional kelp coverage from pyMESMA and spatially degraded CDFW data across nine years revealed relatively low producer's accuracy in detecting kelp, indicating that 38% of pixels detected as kelp from CDFW data were also classified as kelp by MESMA methods (Table 2; Fig. 5a). However, high producer's accuracy for water and user's accuracy for kelp revealed that pyMESMA rarely misclassified water and that pyMESMA kelp classifications were rarely inaccurate (Table 2). The results indicate that the pyMESMA approach is less sensitive than CDFW, but that overall pyMESMA produces high accuracy with few false positives (Fig. 5a). We also calculated a Matthews Correlation Coefficient (MCC) between CDFW and pyMESMA matchups (using the original, non-thinned imagery) of 0.44, which would correspond to approximately 72% of cases predicted accurately for a dataset with balanced positive and negative class sizes (Vihinen, 2012).

Uncertainty statistics between the CDFW record and the LS-4, −5, and − 8 sensors show that kelp coverages obtained from LS-8 satellite images produced much lower RMSE values than those produced from LS-4 and -5 images (Table 3), perhaps due in part to the lower average bull kelp coverage during the LS-8 observational period. NRMSE values were higher and R^2 values were lower for LS-8 compared with LS-4 and -5, most likely because of the limited data availability for LS-8 images and because Landsat tends to underestimate CDFW values in the lower spatial coverage conditions present during the LS-8 period (e.g. Fig. 5). Bell et al. (2020) also found that differences in the spectral band response functions between Landsat sensors decreased the detection of giant kelp by LS-8 relative to LS-4 and -5, but that inter-sensor differences could be mitigated using a quadratic function. A similar procedure might improve the accuracy for bull kelp estimation using LS-8, but was not evaluated here in part due to limited availability of cloud-free imagery for inter-sensor matchup comparisons.

3.3. Tidal analysis

We compared tidal height to canopy estimation from all 88 images from across our timeseries and observed a non-significant positive relationship across an intermediate tidal range (R^2 = 0.072), with variance in the dataset increasing at heights above one meter (p = 0.92). We also found a non-significant positive relationship (Fig. 6; p = 0.27) between tidal height and the residuals of canopy area derived from the pyMESMA/CDFW comparison (Fig. 5c) and a low R^2 value (0.17). The non-significant relationship between tidal height and the residual of canopy area may be part of an underlying pattern caused by changes in the depth of kelp blades from changes in current speed, or increases of foam or white-capping in low-tide imagery. While this should be considered when using satellite-based approaches, the data evaluated herein did not indicate a statistically significant relationship between tidal height and bull kelp detection. Attributing a functional response to tidal height would presumably require a much larger dataset to determine whether this positive relationship is spurious or real.

3.4. Climatological analysis

Development of kelp climatology revealed that canopy area peaked in early fall months, with the largest average areas recorded in August,

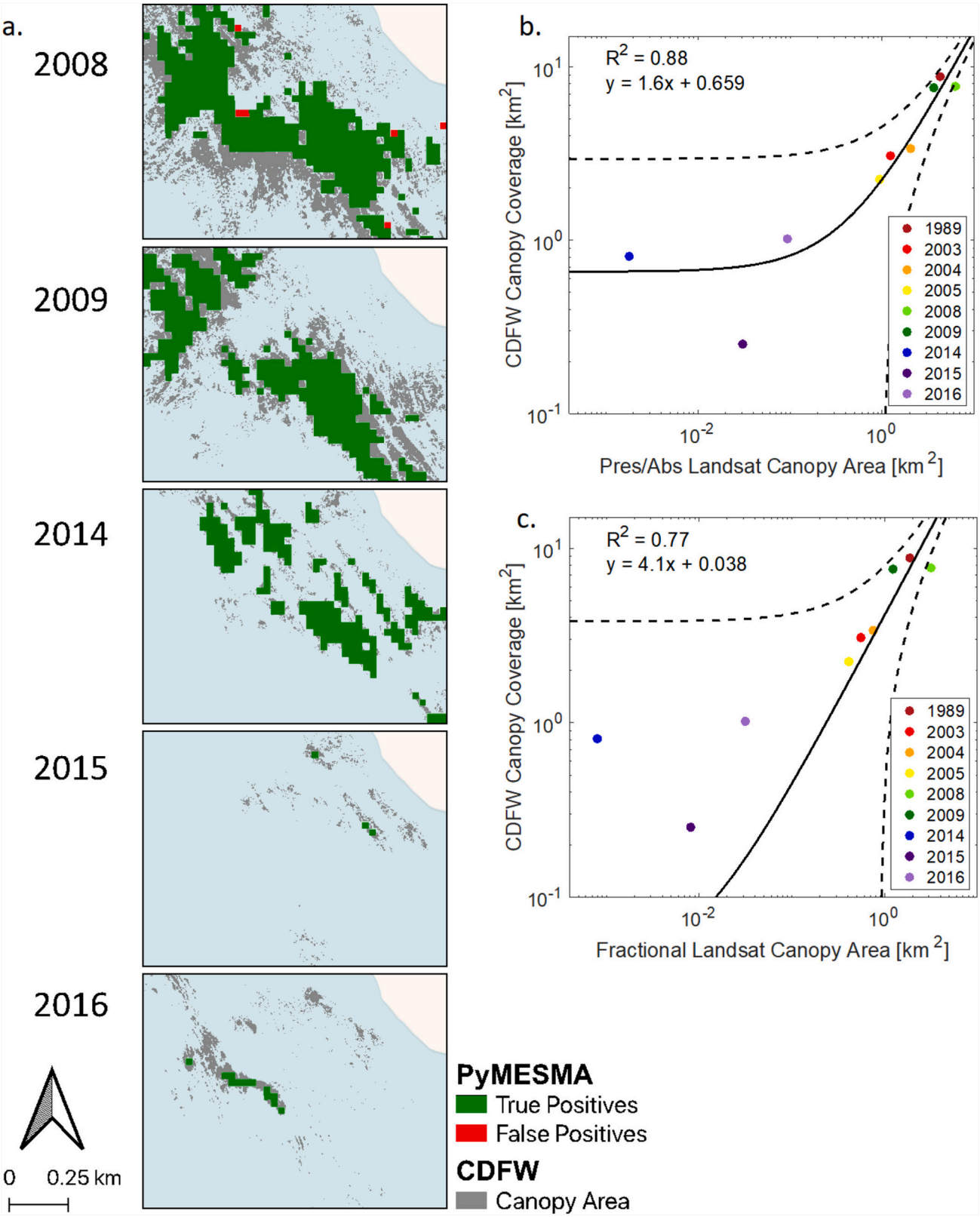


Fig. 5. a. Timeseries comparison of 5 representative years for CDFW and pyMESMA presence/absence matchups of kelp coverage for Point Arena, California (also see Fig. 2). The map compares canopy changes between 2008, 2009, and the recent decline of 2014–2016. True Positives are pyMESMA kelp pixels also detected as kelp by aerial surveys, False Positive pixels represent kelp detected by pyMESMA but not CDFW, and areas with visible CDFW canopy represent 'False Negatives' that pyMESMA did not detect (created in QGIS Version 3.10). b. Matchup between CDFW and pyMESMA canopy area using presence/absence data. c. Matchup using pyMESMA fractional data. Dashed lines in panels b and c indicate ±1 RMSE in the pyMESMA estimate.

Table 1
Comparison of Root-Mean-Square-Error (RMSE), Normalized RMSE (NRMSE), R^2, and relative processing time for different fractional MESMA trials.

Kelp endmember used	MESMA method	RMSE (km^2)	NRMSE	R^2	Relative processing time
Giant Kelp	7-EM Optimized MESMA	1.526	17.87%	0.7591	3.5×
Bull Kelp	7-EM Optimized MESMA	1.444	16.91%	0.7844	3.5×
Bull Kelp	30-EM Scripted pyMESMA	1.507	17.64%	0.7652	1×

Table 2
Error matrix from spatial analysis of CDFW resampled data and pyMESMA results, with cells showing the number of pixels detected as kelp or water by each method (i.e., the cell with CDFW kelp and Landsat kelp depicts pixels detected as kelp by both methods). PA = producer's accuracy; UA = user's accuracy; TA = total accuracy.

Landsat	CDFW			
	Kelp	Water		UA
Kelp	339	6		0.9826
Water	561	894		0.6144
PA	0.3767	0.9933		
			TA:	0.6850

Table 3
Matchup statistics for pyMESMA prediction of kelp coverage for various Landsat sensors using the mean coverage, RMSE, NRMSE, and R^2 values of LS-8 derived coverage (from 2014 to 2016) and LS-4 and -5 derived coverage (1989, 2003 to 2005, 2008, and 2009).

Satellite subset	Mean coverage (km^2)	RMSE (km^2)	NRMSE	R^2
LS-8	0.01344	0.2740	35.68%	0.2845
LS-4 and-5	1.328	1.617	24.68%	0.6188

September, and October (Fig. 7a). Results showed area beginning to decline in November and remaining relatively low for the next eight months, though coverage appeared to reach a winter maximum in February, likely due to anomalous patterns of kelp overwintering.

Individual months from a dataset collected throughout the entire 35-year timeseries indicated high monthly variability in remotely sensed canopy area, with highest coverage occurring in the fall months that precede the arrival of winter storms. Inter-annual variability was high, and comparison of same-year data (Fig. 7 panels b-d) suggest that the optimal image acquisition occurred for September and October retrievals, with August and November providing decreased canopy cover estimates. November coverage in particular showed high variability (indicated by the standard error) as a fraction of the coverage, likely due to variability in the timing of winter storms.

4. Discussion

Our comparison of methodologies for detecting bull kelp indicates that multiple satellite methods enable construction of useful bull kelp canopy observations, with some notable potential optimizations for image selection and processing decisions. In conducting the MESMA process, we found that optimizing water endmembers with EMC to select the seven most representative water endmembers produced the lowest uncertainties and was a useful approach when satellite records contained a smaller number of usable images that were processed manually. Optimization of the water endmembers was time-intensive, however, and since we found that neither the number of endmembers nor the use of a scripted method (pyMESMA) resulted in statistically different outcomes, we chose to use the scripted process for building the bull kelp timeseries. We also found that a four-band bull kelp endmember developed with Landsat pixels containing bull kelp performed better than repurposing an existing giant kelp endmember used previously by Cavanaugh et al. (2011) and others.

The differences in kelp area measured using the Landsat and CDFW approaches were greatest during years of low bull kelp coverage, and the approaches were more comparable during years with higher kelp coverage. While the use of presence/absence criteria can minimize the differences between the Landsat and aerial datasets, the MESMA fractional coverage results indicate a broader range of kelp canopy coverage, with reduced areas during 2014–2016 that are consistent with the expected scope of the bull kelp die-off (McPherson et al., 2020; Renshaw et al., 2017; Rogers-Bennett and Catton, 2019). Landsat MESMA results produced less kelp coverage than CDFW methods, illustrated by error matrix results showing low producer's accuracy in classifying pixels identified as kelp by CDFW. The confusion matrix results are potentially misleading in their presentation of the remote sensing (Landsat MESMA) performance, however, because they treat CDFW data as reference or field data. Thus, inconsistencies in the CDFW dataset, such as methodological changes from manual to automated kelp clustering and assumptions of 100% kelp within a polygon (Veisze et al., 1999), degrade the statistics presented here for the MESMA approach. Comparing total areas, rather than individual pixels, can minimize the effects of these inconsistencies while improving the annual record of bull kelp abundance versus decline across a wide spatial scale. CDFW surveys contain several historical gaps and cannot be conducted annually without increases in allocation of monitoring resources (Jensen and Estes, 1980; Veisze et al., 1999), while free and publicly available Landsat data enable fractional coverage estimates and allow regular remote monitoring across a larger spatial scale using the methods described. Despite methodological differences, the good correspondence between datasets (evidenced by low RMSE and high correlation for the pyMESMA approach), suggests that either dataset can be used to detect spatial and temporal patterns in bull kelp coverage valuable to managers and scientists. The effect of class imbalance on the confusion matrix results was considered by thinning the datasets, as well as by evaluating MCC, which produced values that were consistent with those reported for estimation of giant kelp by optimizing citizen science classifications of Landsat imagery (Rosenthal et al., 2018).

PyMESMA results yielded lower RMSE values for LS-8 images compared to LS-4 and -5, but LS-4 and -5 resulted in lower NRMSE and higher R^2. LS-8 validation statistics were impacted by lower annual kelp coverage, a smaller temporal range, fewer available matchups, and potentially by differences in spectral response functions (Bell et al., 2020), all of which likely influenced the statistical comparison between LS-8 and CDFW coverage. Fig. 5b and c illustrate greater differences between CDFW and Landsat estimates for years with low kelp coverage, perhaps indicating that LS-8 statistics are degraded by differences in detection and response compared with the CDFW record and making it difficult to reach definitive conclusions on accuracy across sensors. LS-4 and -5 satellites utilized Thematic Mapper sensors, but LS-8 implemented Operational Land Imagery sensors with an increased signal-to-noise ratio (Bell et al., 2020), which is anticipated to be helpful for detection of kelp because of the difference in signal at near-infrared wavelengths between kelp and water targets. Bell et al. (2020) observed that the OLI sensor underestimated canopy of giant kelp, which might suggest similar underestimations of bull kelp canopy. Direct comparison and interpretation are difficult, however, without conducting an inter-sensor matchup.

Bell et al. (2020) reported that tides had a significant impact on Landsat detection of giant kelp. Here we found that the residuals in our remote estimation of canopy were not significantly related to tidal cycles, although the number of matchups was small. Considering an

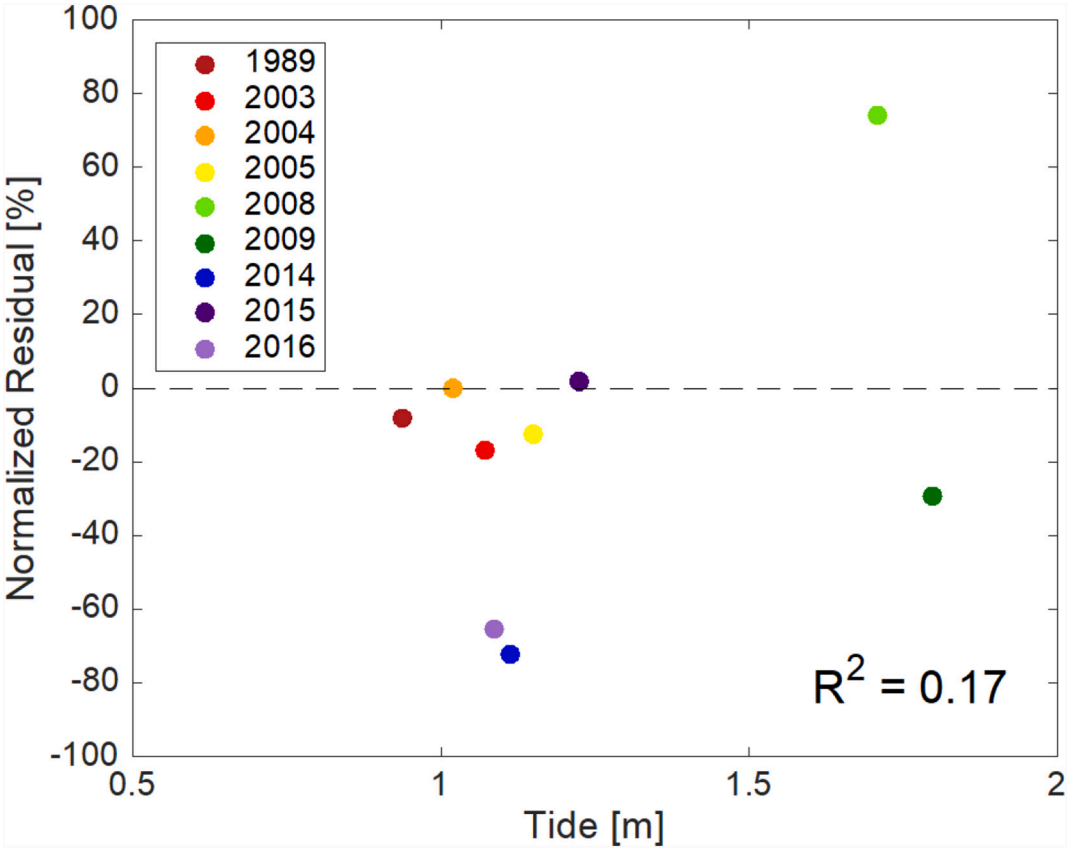


Fig. 6. Comparison of residuals for nine scenes derived from the pyMESMA/CDFW comparison (Fig. 5c) against Point Arena tidal height at time of Landsat imagery.

expanded dataset of pyMESMA canopy estimates, we observed an increase in the variance of the satellite estimation during high tide conditions, but the relationship between tidal cycle and canopy area detection was not significant within an intermediate tidal range. This may suggest that tides are more consequential in remote sensing of giant kelp canopy compared to bull kelp canopy, although we were able to assess fewer matchups compared to Bell et al. (2020).

Different effects of tidal cycles between the two kelp genera may hint at the possible role of morphological differences relevant to remote sensing. A plausible explanation for decreased sensitivity of tidal cycle to remote sensing of bull kelp is that while giant kelp blades float largely at or above the surface (Jensen and Estes, 1980), bull kelp blades often remain submerged (Britton-Simmons et al., 2008; Koehl, 1984; Schroeder et al., 2019), and thus the remote sensing signal contributed by the blades might not change as dramatically across tidal cycles. While CDFW surveys have differentiated canopy and subsurface bull kelp since 2008, MESMA methods and Landsat resolution limit direct comparison (Hamilton et al., 2020). Future remote efforts with improved spatial and temporal resolution might further investigate the role of tides in satellite detection of kelp canopy area within a single year.

Since bull kelp is an annual species unlike the perennial giant kelp (Springer et al., 2010), its coverage varies greatly throughout the year. Thus, survey methods have a narrower window to detect peak biomass, and frequent revisit-time by satellite platforms are conducive to bull kelp surveys. Our results indicate maximum bull kelp coverage from August to October, consistent with publications that describe biomass for bull kelp and other kelp genera peaking in late summer and early autumn (Denny et al., 1997; Huovinen et al., 2020; Schroeder et al., 2019). Comparison of images within individual years suggested that bull kelp coverage increased from August into September and October, and decreased from October into November, making September and October images ideal for remote sensing, although a greater number of years should be compared in future work given the high variability of canopy area (Fig. 5a). Based on the results shown here, bull kelp remote sensing could prioritize image acquisition for September and October, which may require the addition of alternate compatible platforms for years when clear images from this time span are unavailable. Despite the potential of winter storms to remove kelp, February corresponded with relatively high average coverage, indicating that with the right conditions and absence of strong storms, some canopy area may persist (Springer et al., 2010). Further work might investigate this overwintering and variability in the peak coverage by analyzing images throughout the course of a year.

Our analysis targeted scenes from northern California based on the availability of in situ data, but our goal is that a similar approach would be applicable across broader regions. Future research efforts should focus on remotely mapping bull kelp over a broader area of the northeastern Pacific coastline, such as recent work mapping global expanses of giant kelp with Sentinel-2 satellites (Mora-Soto et al., 2020). Although this study did not take into account regionally specific challenges in kelp remote sensing and the timeseries analysis was limited to the period after Sentinel-2 became operational in late 2015, it provided a framework for other methods of kelp detection at large temporal and spatial scales. Many satellites available to monitor coastal ecosystems, such as Sentinel-2, may quantify kelp at improved spatial resolutions across ecologically and climatologically relevant time frames. If low canopy biomass conditions continue to persist in northern California, understanding detection capabilities across a broad range of sensor types will be necessary to effectively monitor kelp ecosystems threatened by climate-driven processing and anthropogenic influences.

5. Conclusions

We demonstrate the utility of publicly available datasets from state

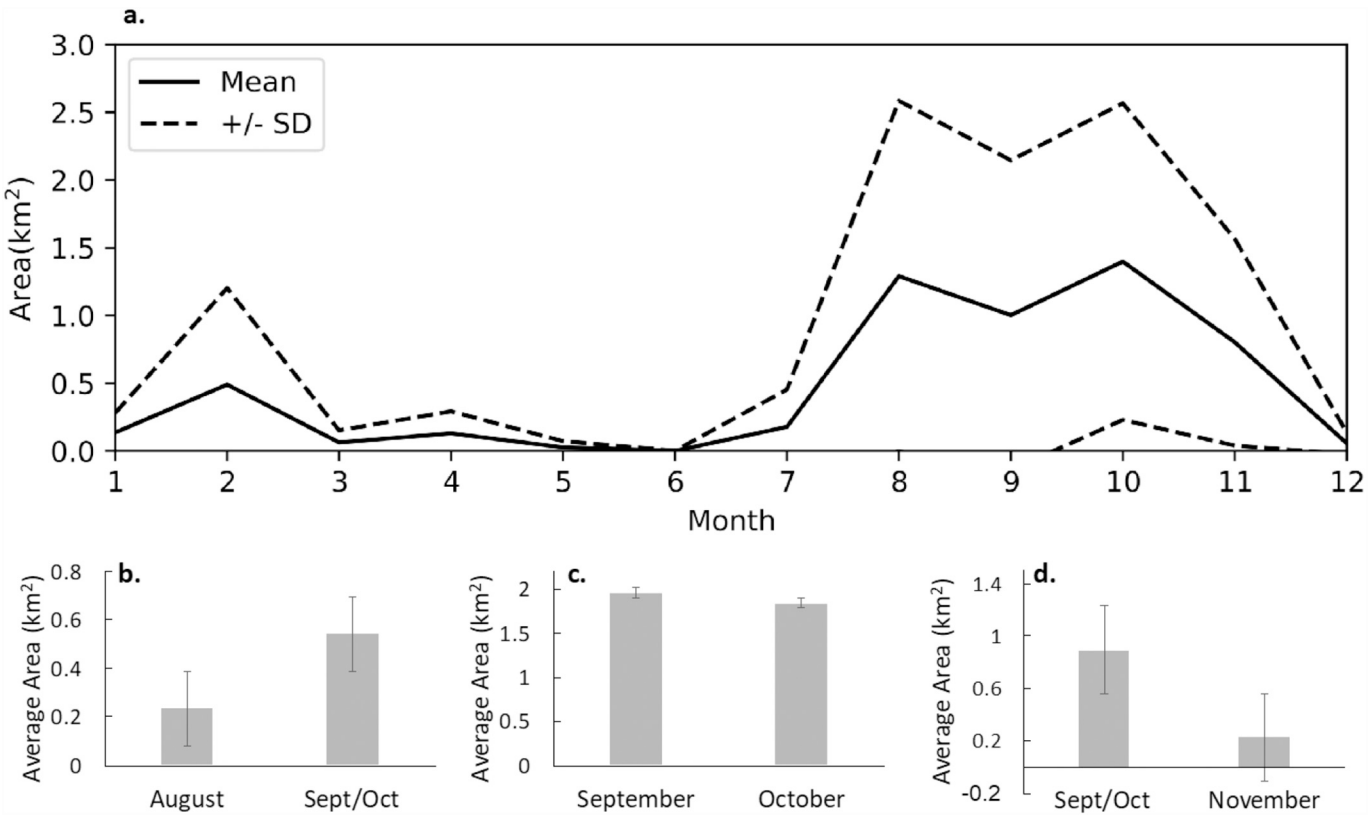


Fig. 7. a. Monthly climatology derived from a 35-year analysis of Landsat imagery using pyMESMA methods to determine timing of maximum kelp coverage. b-d. Comparison of kelp area when multiple scenes from the same year were available. b. August versus combined September/October; data from 1997 and 2018. c. September versus October; data from 2002 and 2016. d. Combined September/October versus November; data from 2002, 2003, 2010, 2014–2016.

and federal agencies to conduct cost-effective monitoring and research of bull kelp ecosystems. The use of Landsat and MESMA enables detection of bull kelp across decadal timespans and produces canopy area estimates comparable to those of aerial surveys despite differences in resolution and methodology. Optimizing water endmembers for MESMA yielded the highest accuracy, but the pyMESMA method performed well while minimizing processing time, which will become increasingly important as the spatial and temporal extent of analyses grow. We observed that LS-8 results had higher NRMSE values than those from LS-4 and -5, but this should be interpreted cautiously given the detection challenges during conditions of diminished kelp cover, which were present throughout the LS-8 monitoring period. Furthermore, we found that tidal height had no significant impact on canopy cover estimates, and bull kelp detection was greatest during the peak months of September and October.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.rse.2020.112243.

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